

User Guide

Rugby Performance Analysis Platform | GPS, physical load and tactical intelligence

This tool turns athletes' **GPS and inertial sensor** data (collected via Catapult Connect) into **clear, actionable insights** on physical performance and tactical behaviour. In just a few clicks you go from raw spreadsheets to **heat maps, load profiles, worst-case scenarios (WCS) and collective tactical visualisations** — including a 3D replay of the match. The goal: to give coaching and performance staff a **fast, visual and scientifically grounded** reading of each athlete and of the team as a system.

Getting started in 4 steps

1

Connect

In the sidebar, enter your Connect **Token** and load the athletes and activities.

2

Choose the activity

Select the **activity** (training or match) you want to analyse from the loaded list.

3

Select period(s) and athletes

Pick one or more **periods** (1st half, 2nd half, etc.) and the **athletes** of interest. With several periods, the tool automatically creates a combined view.

4

Explore the 5 tabs

Data is organised into five modules. Each tab has interactive charts (hover for details, drag to zoom) and export buttons.

The 5 platform modules

SUMMARY

- **Overview:** total distance, PlayerLoad, top speed and key session indicators.
- **By Position:** compares athletes by tactical role (prop, lock, centre, wing...).

FIELD & GPS

- **Rugby Field:** occupancy heat maps and trajectory — this is where you mark the field.
- **Match Story:** heat maps by phase, showing how presence evolves over time.
- **WCS:** the most demanding passages of the match by rolling window.

PHYSICAL LOAD

- **Efforts:** accelerations, decelerations and sprints, with zone and direction.
- **Time Windows:** rolling-window intensity and high / medium-high peaks.
- **Neuromuscular, Acc-Vel and HR:** fatigue, acc×speed profile and cardiac load (TRIMP + zones).

COLLECTIVE TACTICS

- The team as a **system:** cross-references every athlete's position at the same instant.
- Four views: **Pitch Control, Team Breathing, Voronoi** and **3D Replay**.
- Works even with **GPS-only** devices (reconstructs the coordinates).

Marking the field: enabling location data

Raw GPS is just a cloud of **latitude/longitude** points. For the app to understand *where* on the pitch each point is — and thus produce zone occupancy, field thirds, directional efforts and the whole **Collective Tactics** module — you must tell it **where the field is and how it is oriented**. This is called **marking (or calibrating) the field**, and it is done once per venue, over a satellite image.

Where it is

Tab **Field & GPS > Rugby Field**. Select a **reference athlete who has GPS** in the session (the field belongs to the venue, not the athlete — anyone with signal works). If the session has GPS points, a **satellite map** appears with the point cloud and a **yellow target** marking the field centre.

Step-by-step marking

1

Show the field

Click **Show Field** to overlay the field drawing (white lines) on the satellite image.

2

Centre it

Drag the **yellow target** — or edit **Lat/Lon** — until the drawing covers the GPS point cloud, over the real pitch in the image.

3

Rotate

Use the **Rot** buttons (« =-10°, < =-1°, > =+1°, » =+10°) until the **drawing's lines match the real lines** of the field on the satellite. This is the most important step for tactical fidelity.

4

Adjust dimensions

Set **Length** (90–100 m) and **Width** (60–70 m) to match the real size. Default is 100×70 m. **In-goal** sets the depth of the in-goal areas (default 10 m).

5

Apply

Click **Apply Field**. From then on, location coordinates feed the heat maps, directional efforts and the whole **Collective Tactics** module.

Control in the map panel	What it does
Show Field	Toggles the field drawing over the satellite.
Lat / Lon	Moves the field centre. Drag the yellow target or edit the numbers.
Rot « < > »	Rotates the field: « -10°, < -1°, > +1°, » +10°. Align the drawing's lines with the pitch.
Length / Width	Length (90–100 m) and width (60–70 m). Default 100×70 m.
In-goal	Extra margin drawn outside the lines (0–15 m).
Apply Field	Confirms the setup and loads the field coordinates.

Save for everyone (venue database)

Right after applying, the app offers to **save the configuration to the shared venue database**, with a name (e.g. “Municipal Stadium”). It's well worth it: next time **any user** opens an activity **at this same venue**, the field is already **configured automatically** — no need to mark it again. You'll see the notice “Field loaded from the shared database”.

- **Readjust**: redo the marking from scratch (back to the satellite map).
- **Venues**: opens the manager to list, update or delete saved venues.

How to tell it's well marked

- The **white drawing lines** match the field lines in the satellite image.
- The **GPS point cloud** stays inside the field, without spilling far outside.
- The **H-posts** sit on the try lines and the halfway line crosses the real centre of the pitch.
- Tip: start with **rotation**, then adjust the **centre**, and finally the **dimensions**.

Spotlight: the Collective Tactics tab

It's the most innovative module. By syncing the position of all selected athletes, it reveals the team's behaviour as a whole — impossible to see athlete by athlete. It requires **at least 2 athletes with position** in the same period. The four views:

Pitch Control

- William Spearman's model (Liverpool FC). Colours the field by **space dominance** — who would reach each point first, accounting for speed and direction.

Team Breathing

- Shows the block **compressing** when defending and **expanding** in possession, with width, length, area and spread over time.

Voronoi

- The tactical “stained glass”: each region of the field belongs to the nearest player, exposing **space coverage and gaps** in the structure.

3D Replay

- A navigable **synthetic broadcast** (rotate the camera, zoom) with striped grass, goals and markings — reconstructed from coordinates alone.

Animation controls

- **Analysis window**: from 30 s to 10 min, a custom duration or the **whole period**. Short windows show continuous movement (real time); the whole period gives the macro view of the entire match (with more spaced-out frames).
- **Window start**: drag it to place the analysis at **any moment** of the match (including the final minutes).
- **Speed**: **1× = real time** of the match; below is slow motion, above is sped up. Use **Play/Pause** and the slider to navigate the frames.

Two time controls — don't confuse them

- The **Window start** slider (above the chart) **moves the window** across the whole match.
- The slider **inside the chart** only scrubs the **frames of the current window**.

Troubleshooting (common difficulties)

P: The map doesn't appear / "No real GPS point found".

R: That athlete's sensor didn't get a GPS *lock* in the session. Since the field belongs to the venue, pick **another reference athlete** with GPS signal to do the marking.

P: The field is crooked or misaligned on the satellite.

R: First adjust the **rotation** (« < > ») and then the **centre** (Lat/Lon) until the drawing's lines match the pitch. Also check **Length/Width** — wrong dimensions distort everything.

P: Collective Tactics shows "coordinates reconstructed from GPS / approximate alignment".

R: Your device is **GPS-only** (no native x/y). The **relative positions** between players are faithful; for better absolute alignment, **mark the field** in Field & GPS — it will be used automatically here.

P: "This tab needs at least 2 athletes with position in the same period".

R: In the sidebar, select **2 or more athletes** that have GPS in that period.

P: The animation looks like "photos" jumping, with no continuous motion.

R: Long passages have spaced-out frames. Reduce the **Analysis window** to **1–2 min** and use the **Window start** to focus on the play. The whole period is for the macro view.

P: GPS distances differ from what I expected.

R: Check that the field is well marked and with the **real dimensions** of the pitch. The app also compares the field-filtered distance with the integrated GPS — big differences point to a marking that needs review.

P: I configured the field but want to redo it.

R: Use **Readjust** (back to the map) or **Venues** to update/delete the saved venue.

Tips to get the most out of it

- **Save the venue** after marking the field: it benefits the whole team automatically.
- **Combine periods** to see the whole match on a continuous timeline, without half-time.
- **Export** tables and events to CSV for reports and longitudinal tracking.
- To study a play in real time, use a **1–2 min** window and aim the start at the exact moment.
- Hover over the charts to see **exact values**; drag to zoom.

Additions, suggestions, comments

Have ideas to improve the tool, found something to adjust, or want to suggest a new feature? Your feedback is very welcome. Write to:

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